# Dr. Ryan Chown

Assistant Professor
Faculty of Computer Science and Technology, Algoma University
WW 105C, 1520 Queen Street East, Sault Ste. Marie, Ontario Canada, P6A 2G4
ryan.chown@algomau.ca ⧫ (905) 520 9634
`https://rchwn.github.io`

## Interests and expertise

The interstellar medium and star formation in galaxies. Application of mathematical and computational techniques to astronomical research and large datasets. Infrared and radio observations with JWST, ALMA and the VLA.

## Education

**McMaster University** — Hamilton, ON
Ph.D. Astronomy — 08/2021
**Thesis:** *Multi-wavelength studies of the interstellar medium and star formation in nearby galaxies*
**Advisors:** Prof. Christine D. Wilson and Prof. Laura Parker
Finalist, J. S. Plaskett Medal for Most Outstanding PhD Thesis, Canada

**McGill University** — Montreal, QC
M.Sc. Physics — 05/2015 - 05/2017
**Thesis:** *Mapping the Millimeter-wave Sky with Combined South Pole Telescope and* Planck *Data*
**Advisor:** Prof. Gil Holder

**McGill University** — Montreal, QC
B.Sc. Physics, with a Minor in Computer Science — 09/2012 - 04/2015

## Academic Positions

**Algoma University** — Sault Ste. Marie, ON
Assistant Professor (Tenure-Track) — 9/2025-
**The Ohio State University** — Columbus, OH
Postdoctoral Scholar — 10/2023-9/2025
**The University of Western Ontario** — London, ON
Postdoctoral Scholar — 10/2021-09/2023

## Grants

NSERC Discovery Grant, $195,000 CAD over 5 years — 5/2026
NSERC Discovery Launch Supplement, $12,500 CAD — 5/2026

## Teaching Experience

PHYS 1906 Instructor — Algoma
*General Astronomy I* — S26
MATH 1036 Instructor — Algoma
*Calculus I* — S26
MATH 2056 Instructor — Algoma

| | |
|---|---|
| *Discrete Mathematics II* | W26 |
| PHYS 1007 Instructor | Algoma |
| *Introductory Physics II* | W26 |
| PHYS 1006 Instructor | Algoma |
| *Introductory Physics I* | W25 |
| PHYS 1AA3 Teaching Assistant | McMaster |
| *Introduction to Modern Physics* | 2020 |
| PHYS 1A03 Teaching Assistant | McMaster |
| *Introductory Physics* | 2018, 2019, 2020 |
| PHYS 1E03 Teaching Assistant | McMaster |
| *Waves, Electricity, and Magnetic Fields* | 2018 |
| ASTRO 1F03 Teaching Assistant | McMaster |
| *Astronomy and Astrophysics* | 2017, 2021 |
| ASTRO 2C03 Teaching Assistant | McMaster |
| *Big Questions* | 2017, 2021 |
| PHYS 183 Teaching Assistant | McGill |
| *The Milky Way and Beyond* | 2016 and 2017 |
| PHYS 230 Teaching Assistant | McGill |
| *Dynamics of Simple Systems* | 2015 and 2016 |

## Research Supervision Experience

Notes: ⋆ indicates that I was the primary (informal) supervisor
† = student held a MITACS Globalink Research Award

5. ⋆,†Rajarshi Choudhury, *Indian Inst. of Sci. Ed. and Research Bhopal (undergraduate)* 2023
4. †Arnab Chowhan, *Centre for Excellence in Basic Sciences, Mumbai (undergraduate)* 2023
3. ⋆,†Raphaela Kestler, *Durham University (undergraduate)* 2022
2. ⋆,†Holly Raynor, *Durham University (undergraduate)* 2022
1. ⋆Khaleda Ramzi, *Western University Work-Study Program (undergraduate)* 2022

## Service

| | |
|---|---|
| External Panelist for JWST Cycle 5 Proposals | 2025 |
| Referee for ApJ, A&A, MNRAS | 2023– |

## Professional Development

Western Certificate for University Teaching and Learning — January 2024
- An intensive five-component program designed "to enhance the quality of teaching by graduate students and postdoctoral scholars, and to prepare them for a future faculty or professional career"
- `https://teaching.uwo.ca/programs/certificates/cutl.html`

"What makes a great postdoc mentor" – Workshop at UWO — January 2022

## Publications

*Overall summary:* ***87*** *peer-reviewed papers;* ***2412*** *citations; h-index=**30***

**A. ≤ 3rd author papers**

*Summary:* ***11*** *papers,* ***386*** *citations as of June, 2026.*

11. Sutter, J., Sandstrom, K., **Chown, R.**, *et al.*, *Characterization of Two Cool Galaxy Outflow Candidates Using Mid-infrared Emission from Polycyclic Aromatic Hydrocarbons*. ApJL 992 (2025): L7

10. **Chown, R.**, *et al.*, *Relationships between Polycyclic Aromatic Hydrocarbons, Small Dust Grains, $H_2$, and H I in Local Group Dwarf Galaxies NGC 6822 and WLM Using JWST, ALMA, and the VLA*. ApJ 987 (2025): 91

9. **Chown, R.**, *et al.*, *PDRs4All: XIII. Empirical prescriptions for the interpretation of JWST imaging observations of star-forming regions*. A&A 698 (2025): A86

8. **Chown, R.**, *et al.*, *Polycyclic Aromatic Hydrocarbon and CO(21) Emission at 50150 pc Scales in 70 Nearby Galaxies*. ApJ 983 (2025): 64

7. **Chown, R.**, *et al.*, *PDRs4All. IV. An embarrassment of riches: Aromatic infrared bands in the Orion Bar*. A&A 685 (2024): A75

6. **Chown, R.**, *et al.*, *The cold gas and dust properties of red star-forming galaxies*. MNRAS 516 (2022): 84-99

5. **Chown, R.**, *et al.*, *A new estimator of resolved molecular gas in nearby galaxies*. MNRAS 500 (2021): 1261-1278

4. **Chown, R.**, *et al.*, *Linking bar- and interaction-driven molecular gas concentration with centrally enhanced star formation in EDGE-CALIFA galaxies*. MNRAS 484 (2019): 5192-5211

3. **Chown, R.**, *et al.*, *Maps of the Southern Millimeter-wave Sky from Combined 2500 deg$^2$ SPT-SZ and Planck Temperature Data*. ApJS 239 (2018): 10

2. Omori, Y., **Chown, R.**, *et al.*, *A 2500 deg$^2$ CMB Lensing Map from Combined South Pole Telescope and Planck Data*. ApJ 849 (2017): 124

1. Crawford, T., **Chown, R.**, *et al.*, *Maps of the Magellanic Clouds from Combined South Pole Telescope and PLANCK Data*. ApJS 227 (2016): 23

**B. Other papers with significant contributions during or after PhD**
*Summary: **59** papers, **1193** citations as of June, 2026.*

59. Hassani, H., *et al.*, *The Hidden Life of Stars: Embedded Beginnings to Asymptotic Giant Branch Endings in the PHANGSJWST Sample. I. Catalog of Mid-infrared Sources*. ApJS 284 (2026): 3

58. Khan, B., *et al.*, *PDRs4All: XX. The 69 μm region as a probe of PAH charge and size in the Orion Bar*. A&A 709 (2026): A39

57. Maragkoudakis, A., *et al.*, *PDRs4All: XIX. The evolution of the PAH ionisation and PAH size distribution across the Orion Bar*. A&A 709 (2026): A38

56. Bazzi, Z., *et al.*, *The lifetime of 100,000 molecular clouds in the nearby Universe*. arXiv e-prints (2026): arXiv:2604.24759

55. Kim, J., *et al.*, *Localized Deviations from the COPolycyclic Aromatic Hydrocarbon Relation in PHANGS-JWST Galaxies: Faint Polycyclic Aromatic Hydrocarbon Emission or Elevated CO Emissivity?*. ApJ 1001 (2026): 67

54. Lopez, S., *et al.*, *JWST Observations of Starbursts: Polycyclic Aromatic Hydrocarbons Closely Trace the Cool Phase of M82's Galactic Wind*. ApJL 999 (2026): L7

53. McClain, R., *et al.*, *Resolved H II Regions in NGC 253: Ionized Gas Structure and Suggestions of a Universal DensitySurface Brightness Relation*. ApJ 998 (2026): 166
52. Ramambason, L., *et al.*, *Duration and properties of the embedded phase of star formation in 37 nearby galaxies from PHANGS-JWST*. A&A 706 (2026): A186
51. Bazzi, Z., *et al.*, *PHANGS-JWST: The largest extragalactic molecular cloud catalog traced by polycyclic aromatic hydrocarbon emission*. A&A 706 (2026): A40
50. Zannese, M., *et al.*, *PDRs4All: XVII. Formation and excitation of HD in photodissociation regions: Application to the Orion Bar*. A&A 705 (2026): A128
49. Tarantino, E., *et al.*, *JWST Captures Growth of Aromatic Hydrocarbon Dust Particles in the Extremely Metal-poor Galaxy Sextans A*. arXiv e-prints (2025): arXiv:2512.04060
48. Sun, J., *et al.*, *Resolved Profiles of Stellar Mass, Star Formation Rate, and Predicted CO-to-$H_2$ Conversion Factor Across Thousands of Local Galaxies*. ApJ 994 (2025): 263
47. Graham, G., *et al.*, *PAH Marks the Spot: Digging for Buried Clusters in Nearby Star-forming Galaxies*. AJ 170 (2025): 340
46. Pathak, D., *et al.*, *Masses, Star Formation Efficiencies, and Dynamical Evolution of 18,000 H II Regions*. ApJL 993 (2025): L20
45. Oakes, E., *et al.*, *The Hierarchical Dynamical State of Molecular Gas from 3 to 300 pc in NGC 253*. ApJ 993 (2025): 193
44. Egorov, O., *et al.*, *Polycyclic aromatic hydrocarbon destruction in star-forming regions across 42 nearby galaxies*. A&A 703 (2025): A103
43. Koch, E., *et al.*, *The Karl G. Jansky Very Large Array Local Group L-Band Survey (LGLBS)*. ApJS 279 (2025): 35
42. Sarbadhicary, S., *et al.*, *A First Look at Spatially Resolved Infrared Supernova Remnants in M33 with JWST*. ApJ 989 (2025): 138
41. Kim, J., *et al.*, *Timescales of Polycyclic Aromatic Hydrocarbon and Dust Continuum Emission from Gas Clouds Compared to Molecular Gas Cloud Lifetimes in PHANGS-JWST Galaxies*. ApJ 988 (2025): 215
40. Khan, B., *et al.*, *PDRs4All: XIV. Probing CH out-of-plane bending modes of PAH molecules in the Orion Bar with JWST*. A&A 699 (2025): A133
39. Elmegreen, B., *et al.*, *An Investigation of Disk Thickness in M51 from Hα, Paα, and Mid-infrared Power Spectra*. ApJ 986 (2025): 13
38. Leroy, A., *et al.*, *Cloud-scale Gas Properties, Depletion Times, and Star Formation Efficiency per Freefall Time in PHANGSALMA*. ApJ 985 (2025): 14
37. Draine, B., *et al.*, *Detection of Deuterated Hydrocarbon Nanoparticles in the Whirlpool Galaxy, M51*. ApJL 984 (2025): L42
36. Williams, T., *et al.*, *The resolved star-formation efficiency of early-type galaxies*. MNRAS 538 (2025): 3219-3246
35. Pathak, D., *et al.*, *Linking Stellar Populations to H II Regions across Nearby Galaxies. II. Infrared Reprocessed and UV Direct Radiation Pressure in H II Regions*. ApJ 982 (2025): 140

34. Goicoechea, J., *et al.*, *PDRs4All: XII. Far-ultraviolet-driven formation of simple hydrocarbon radicals and their relation with polycyclic aromatic hydrocarbons*. A&A 696 (2025): A100

33. Zannese, M., *et al.*, *PDRs4All: XI. Detection of infrared $CH^+$ and $CH_3{}^+$ rovibrational emission in the Orion Bar and disk d203-506: Evidence of chemical pumping*. A&A 696 (2025): A99

32. Whitmore, B., *et al.*, *Empirical SED Templates for Star Clusters Observed with HST and JWST: No Strong PAH or IR Dust Emission after 5 Myr*. ApJ 982 (2025): 50

31. Dale, D., *et al.*, *PAH Feature Ratios around Stellar Clusters and Associations in 19 Nearby Galaxies*. AJ 169 (2025): 133

30. Chastenet, J., *et al.*, *The Resolved Behavior of Dust Mass, Polycyclic Aromatic Hydrocarbon Fraction, and Radiation Field in ~800 Nearby Galaxies*. ApJS 276 (2025): 2

29. Pingel, N., *et al.*, *The Local Group L-band Survey: The First Measurements of Localized Cold Neutral Medium Properties in the Low-metallicity Dwarf Galaxy NGC 6822*. ApJ 974 (2024): 93

28. Goicoechea, J., *et al.*, *PDRs4All: X. ALMA and JWST detection of neutral carbon in the externally irradiated disk d203-506: Undepleted gas-phase carbon*. A&A 689 (2024): L4

27. Sutter, J., *et al.*, *The Fraction of Dust Mass in the Form of Polycyclic Aromatic Hydrocarbons on 1050 pc Scales in Nearby Galaxies*. ApJ 971 (2024): 178

26. Williams, T., *et al.*, *PHANGS-JWST: Data-processing Pipeline and First Full Public Data Release*. ApJS 273 (2024): 13

25. Mayker Chen, N., *et al.*, *H$\alpha$ Emission and H II Regions at the Locations of Recent Supernovae in Nearby Galaxies*. AJ 168 (2024): 5

24. Fuente, A., *et al.*, *PDRs4All. IX. Sulfur elemental abundance in the Orion Bar*. A&A 687 (2024): A87

23. Van De Putte, D., *et al.*, *PDRs4All. VIII. Mid-infrared emission line inventory of the Orion Bar*. A&A 687 (2024): A86

22. Zannese, M., *et al.*, *OH as a probe of the warm-water cycle in planet-forming disks*. Nature Astronomy 8 (2024): 577-586

21. Schroetter, I., *et al.*, *PDRs4All. VII. The 3.3 μm aromatic infrared band as a tracer of physical properties of the interstellar medium in galaxies*. A&A 685 (2024): A78

20. Pasquini, S., *et al.*, *PDRs4All. VI. Probing the photochemical evolution of PAHs in the Orion Bar using machine learning techniques*. A&A 685 (2024): A77

19. Elyajouri, M., *et al.*, *PDRs4All. V. Modelling the dust evolution across the illuminated edge of the Orion Bar*. A&A 685 (2024): A76

18. Peeters, E., *et al.*, *PDRs4All: III. JWST's NIR spectroscopic view of the Orion Bar*. A&A 685 (2024): A74

17. Habart, E., *et al.*, *PDRs4All. II. JWST's NIR and MIR imaging view of the Orion Nebula*. A&A 685 (2024): A73

16. Berné, O., *et al.*, *A far-ultravioletdriven photoevaporation flow observed in a protoplanetary disk*. Science 383 (2024): 988-992

15. Teng, Y., *et al.*, *Star Formation Efficiency in Nearby Galaxies Revealed with a New CO-to-$H_2$ Conversion Factor Prescription*. ApJ 961 (2024): 42

14. Pathak, D., *et al.*, *A Two-Component Probability Distribution Function Describes the Mid-IR Emission from the Disks of Star-forming Galaxies*. AJ 167 (2024): 39

13. Changala, P., *et al.*, *Astronomical $CH_3^+$ rovibrational assignments. A combined theoretical and experimental study validating observational findings in the d203-506 UV-irradiated protoplanetary disk*. A&A 680 (2023): A19

12. Berné, O., *et al.*, *Formation of the methyl cation by photochemistry in a protoplanetary disk*. Nature 621 (2023): 56-59

11. Roberts, I., *et al.*, *VERTICO. VI. Cold-gas asymmetries in Virgo cluster galaxies*. A&A 675 (2023): A78

10. Jiménez-Donaire, M., *et al.*, *VERTICO. III. The Kennicutt-Schmidt relation in Virgo cluster galaxies*. A&A 671 (2023): A3

9. Leroy, A., *et al.*, *PHANGS-JWST First Results: A Global and Moderately Resolved View of Mid-infrared and CO Line Emission from Galaxies at the Start of the JWST Era*. ApJL 944 (2023): L10

8. Leroy, A., *et al.*, *PHANGS-JWST First Results: Mid-infrared Emission Traces Both Gas Column Density and Heating at 100 pc Scales*. ApJL 944 (2023): L9

7. **Chown, R.**, *et al.*, *Correction to: The cold gas and dust properties of red star-forming galaxies*. MNRAS 518 (2023): 4536-4536

6. Zabel, N., *et al.*, *Erratum: "VERTICO II: How H I-identified Environmental Mechanisms Affect the Molecular Gas in Cluster Galaxies" (2022, ApJ, 933, 10)*. ApJ 942 (2023): 56

5. Gao, Y., *et al.*, *The Correlation between WISE 12 μm Emission and Molecular Gas Tracers on Subkiloparsec Scales in Nearby Star-forming Galaxies*. ApJ 940 (2022): 133

4. Zabel, N., *et al.*, *VERTICO II: How H I-identified Environmental Mechanisms Affect the Molecular Gas in Cluster Galaxies*. ApJ 933 (2022): 10

3. Berné, O., *et al.*, *PDRs4All: A JWST Early Release Science Program on Radiative Feedback from Massive Stars*. PASP 134 (2022): 054301

2. Brown, T., *et al.*, *VERTICO: The Virgo Environment Traced in CO Survey*. ApJS 257 (2021): 21

1. Brown, T., *et al.*, *VERTICO: Virgo Environment Traced In CO*. (2020): 13

### C. Other papers with significant contributions prior to PhD

*Summary: **17** papers, **833** citations as of June, 2026.*

17. Sánchez, J., *et al.*, *Mapping gas around massive galaxies: cross-correlation of DES Y3 galaxies and Compton-y maps from SPT and Planck*. MNRAS 522 (2023): 3163-3182

16. Omori, Y., *et al.*, *Joint analysis of Dark Energy Survey Year 3 data and CMB lensing from SPT and Planck. I. Construction of CMB lensing maps and modeling choices*. Phys. Rev. D 107 (2023): 023529

15. Abbott, T., *et al.*, *Joint analysis of Dark Energy Survey Year 3 data and CMB lensing from SPT and Planck. III. Combined cosmological constraints*. Phys. Rev. D 107 (2023): 023531

14. Chang, C., *et al.*, *Joint analysis of Dark Energy Survey Year 3 data and CMB lensing from SPT and Planck. II. Cross-correlation measurements and cosmological constraints*. Phys. Rev. D 107 (2023): 023530
13. Anbajagane, D., *et al.*, *Shocks in the stacked Sunyaev-Zel'dovich profiles of clusters II: Measurements from SPT-SZ + Planck Compton-y map*. MNRAS 514 (2022): 1645-1663
12. Salvati, L., *et al.*, *Combining Planck and SPT Cluster Catalogs: Cosmological Analysis and Impact on the Planck Scaling Relation Calibration*. ApJ 934 (2022): 129
11. Bleem, L., *et al.*, *CMB/kSZ and Compton-y Maps from 2500 deg$^2$ of SPT-SZ and Planck Survey Data*. ApJS 258 (2022): 36
10. Omori, Y., *et al.*, *Dark Energy Survey Year 1 Results: Cross-correlation between Dark Energy Survey Y1 galaxy weak lensing and South Pole Telescope+Planck CMB weak lensing*. Phys. Rev. D 100 (2019): 043517
9. Omori, Y., *et al.*, *Dark Energy Survey Year 1 Results: Tomographic cross-correlations between Dark Energy Survey galaxies and CMB lensing from South Pole Telescope +Planck*. Phys. Rev. D 100 (2019): 043501
8. Prat, J., *et al.*, *Cosmological lensing ratios with DES Y1, SPT, and Planck*. MNRAS 487 (2019): 1363-1379
7. Mocanu, L., *et al.*, *Consistency of cosmic microwave background temperature measurements in three frequency bands in the 2500-square-degree SPT-SZ survey*. JCAP 2019 (2019): 038
6. Abbott, T., *et al.*, *Dark Energy Survey year 1 results: Joint analysis of galaxy clustering, galaxy lensing, and CMB lensing two-point functions*. Phys. Rev. D 100 (2019): 023541
5. Simard, G., *et al.*, *Constraints on Cosmological Parameters from the Angular Power Spectrum of a Combined 2500 deg$^2$ SPT-SZ and Planck Gravitational Lensing Map*. ApJ 860 (2018): 137
4. Hou, Z., *et al.*, *A Comparison of Maps and Power Spectra Determined from South Pole Telescope and Planck Data*. ApJ 853 (2018): 3
3. Aylor, K., *et al.*, *A Comparison of Cosmological Parameters Determined from CMB Temperature Power Spectra from the South Pole Telescope and the Planck Satellite*. ApJ 850 (2017): 101
2. Baxter, E., *et al.*, *Joint measurement of lensing-galaxy correlations using SPT and DES SV data*. MNRAS 461 (2016): 4099-4114
1. Soergel, B., *et al.*, *Detection of the kinematic Sunyaev-Zel'dovich effect with DES Year 1 and SPT*. MNRAS 461 (2016): 3172-3193

**D. Abstracts in proceedings**

4. **R. Chown**, et al. "Polycyclic Aromatic Hydrocarbon and CO (2-1) Emission at 50-150 pc Scales in 70 Nearby Galaxies." AstroPAH Newsletter, 117 (April 2025).
3. **R. Chown**, et al. "PDRs4All IV. An embarrassment of riches: the Aromatic Infrared Bands in the Orion Bar." AstroPAH Newsletter, 104 (December 2023).
2. **R. Chown**, et al. "The Relationship Between Mid-infrared and CO Emission at kpc Scales in Nearby Galaxies." Bulletin of the American Astronomical Society, Vol. 53, No. 6 (2021).
1. T. Brown, et al. (incl. **R. Chown**) "VERTICO: Virgo Environment Traced In CO." Scientific Meeting of the Spanish Astronomical Society, 13-15 July 2020.

## Public Talks

5. Public Astro Night, Queen's University — Kingston, ON
   Public talk on star formation and galaxies with JWST — 8 June 2024
4. Amica Retirement Home — London, ON
   "Are We Alone?" – a talk about searching for life in the Universe with JWST — June 2023
3. Parkwood Institute — London, ON
   "Are We Alone?" – a talk about searching for life in the Universe with JWST — January 2023
2. JWebbinar 23: PDRs4All Community Telecon
   Gave a public talk on status of JWST observations from PDRs4All — December 2022
1. Astronomy on Tap — Montreal, QC
   "Patterns in the Cosmic Microwave Background" — May 2017

## Outreach

Canadian Bushplane Heritage Centre — Sault Ste. Marie, ON
  Facilitated 2-day "Health in Space" hands-on event for Grade 6 students — March 2026
    CBC Coverage: `https://www.cbc.ca/player/play/video/9.7149669`
Algoma University, Global Game Jam — Sault Ste. Marie, ON
  Co-organizer of the Algoma University Global Game Jam — 2026
STEAM Factory, Ohio State University — Columbus, OH
  Gave a talk on "stellar birth" — 2024
McMaster Undergraduate Physics Society — Hamilton, ON
  Organized and gave a seminar on Python for undergraduates in physics at McMaster — 2021
Astro McGill — McGill University
  Helped facilitate public Astro Nights — 2016-2017
Physics Matters — McGill University
  Helped organize this series of public talks on physics — 2016-2017
Eureka! Science Festival — Montreal, QC
  Instructed physics experiments for elementary school students — 2015

## Invited Seminars

9. Astro Tea talk, Indiana University, Bloomington — 01/2025
8. Astronomy Journal Club, Waterloo Center for Astrophysics — 10/2024
7. Astronomy Seminar, Queen's University Department of Physics & Astronomy — 06/2024
6. PHANGS Colloquium — 02/2024
5. Astronomy Journal Club, McMaster University — 11/2023
4. CANadian Virtual Astronomy Seminar (CANVAS) — 03/2023
3. Ringberg Seminar Series, MPIA, Germany — 05/2021
2. Extragalactic Database for Galaxy Evolution Meeting, University of Maryland — 05/2021
1. Shanghai Astronomical Observatory, Shanghai, China — 08/2018

## Contributed Talks

| | |
|---|---|
| *Star Formation, Stellar Feedback, and the Ecology of Galaxies*, Visegrad, Hungary | 05/2025 |
| *PHANGS Collaboration Meeting*, Nice, France | 02/2025 |
| *The Physics and Impact of Astrophysical Dust: Star Formation to Cosmology*, Aspen, CO | 03/2024 |
| *PHANGS Collaboration Meeting*, Garching, Germany | 02/2024 |
| *Illuminating the Dusty Universe: A Tribute to the Work of Bruce Draine*, Florence, Italy | 10/2023 |
| *Symposium on the Life Cycle of Cosmic PAHs*, Aarhus University, Denmark | 09/2022 |
| *KIAA Forum on Gas in Galaxies for Early Career Scientists*, Peking University, Beijing | 11/2021 |
| *238th AAS Meeting – Dissertation Talk* | 7/2021 |
| *Exploring Gas in and Around Galaxies* meeting, Tsinghua University, Beijing | 7/2018 |
| *KIAA Forum on Gas in Galaxies*, Peking University, Beijing | 6/2018 |
| *SDSS Chinese MaNGA Meeting*, University of Chinese Academy of Sciences, Beijing | 6/2018 |
| *South Pole Telescope Collaboration Conference*, University of Chicago, IL | 7/2017 |
| *The Centre for Research in Astrophysics of Quebec (CRAQ) Annual Meeting*, Montreal, QC | 5/2017 |
| *South Pole Telescope Collaboration Conference*, University of Chicago, IL | 8/2016 |
| *South Pole Telescope Collaboration Conference*, University of Chicago, IL | 7/2015 |

## Conference Posters

| | |
|---|---|
| *First Science Results from JWST*, STScI, Baltimore | 9/2023 |
| *Canadian Astronomical Society Annual Conference*, York University, Toronto, ON | 5/2020 |
| *Views on the ISM in galaxies in the ALMA era*, University of Bologna, Italy | 9/2019 |
| *Canadian Astronomical Society Annual Conference*, McGill University, Montreal, QC | 6/2019 |

## Approved Observing Proposals

**As P.I. or Co-P.I.:**

5. VLA Cycle 26A (co-PI; PI: Eibensteiner, C.) – “High Quality HI Coverage of the Northern PHANGS JWST+HST+ALMA Targets.”

4. ALMA Cycle 11 (2024.1.01179.S; P.I.) – “A Complete View of Low Metallicity Star Forming Complexes in the Local Group Dwarf NGC 6822.”

3. JCMT 21A (P.I.) – “Measuring global CO(2-1) to supplement interferometric observations from the EDGE survey.”

2. JCMT 18B (P.I.) – “Observing red star-forming galaxies from xCOLD GASS with SCUBA-2.”

1. JCMT 18A (P.I.) – “Extending the JINGLE RxA Samples to Include ‘Red Misfit’ Galaxies.”

**As Co-I.:**

18. JWST Cycle 5: “Where the PAH-CO Relation Breaks: JWST Spectroscopy of High CO/PAH Ridges in NGC 1566.” PI: J. Kim.

17. JWST Cycle 5: “Dust in the Deserts: Understanding PAH Emission in Quiescent Bulges and Star-formation Deserts.” PI: D. Pathak.

16. JWST Cycle 5: “ Resolving Stellar Feedback in Action via NIRCam and MIRI Imaging of Very Nearby Galaxies.” PI: A. Leroy.

15. JWST Cycle 5: “A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33.” PI: E. Rosolowsky.
14. JWST Cycle 4: “Exploring the Last Watering Hole of Low Metallicity PAH Emission: Deep Spectroscopic Observations of Sextans A.” PI: L. Tarantino.
13. JWST Cycle 4: “Resolving the thin disk of a nearby Milky Way analog.” PI: A. Leroy.
12. ALMA Cycle 11: “Linking Molecular Cloud Structure to Massive Star Formation: 5000 molecular clouds, filaments, and bubbles across M33.” PI: E. Koch.
11. ALMA Cycle 11: “Completing the Virgo High-resolution CO 2-1 Survey: Dissecting Galaxy Quenching with Molecular Cloud Scale Micro-physics”. PI: J. Sun.
10. ALMA Cycle 11: “The Last Refuge of CO and PAH Emission in the Most Metal Poor Galaxies.” PI E. Tarantino.
9. HST Cycle 32: “Decoding stellar feedback in action with an HST+MUSE+JWST full-disk survey of starburst galaxy prototype NGC 253.” PI D. Thilker.
8. HST Cycle 32: “Bringing HST to the VLA: The Interaction of Stars and Gas in the Local Group.” PI J. Dalcanton.
7. ALMA Cycle 10: “Virgo High-resolution CO(2-1) Survey: Dissecting Galaxy Quenching with Molecular Cloud Scale ’Micro-physics’”. PI: J. Sun.
6. JWST Cycle 3: “A Systematic Study of the 3.3 - 3.5 $\mu$m PAH Features at $z = 0$ with Archival NIRSpec Observations.” PI K. Sandstrom.
5. ALMA Cycle 9 – “The chemical richness of the Orion Bar and its role as a lab of $CH_3CN$ chemistry in disk-evolved systems.” P.I. F. Alarcón.
4. ALMA Cycle 7 (Large Program) – “The Virgo Environment Traced in CO survey (VERTICO).” P.I. T. Brown.
3. ALMA Cycle 7 – “Mapping CO emission in galaxies from the JINGLE survey.” P.I. C.D. Wilson.
2. JCMT 20A (Large Program) – “JINGLE at the edge: the ISM of starbursts and green valley galaxies.” P.I. L.-H. Lin.
1. JCMT 19B – “Observing CO(2-1) in Red Star-forming Galaxies.” P.I. L.-H. Lin.

## Collaborations

6. The Physics at High Angular Resolution in Nearby GalaxieS (PHANGS) Collaboration — 10/2023–
`http://www.phangs.org`
5. The Local Group L-Band Survey Collaboration — 10/2023–
`https://www.lglbs.org`
4. PDRs4All: Radiative Feedback from Massive Stars as Traced by Multiband Imaging and Spectroscopic Mosaics – A JWST Early Release Science (ERS) Program — 10/2021–
`https://pdrs4all.org`
3. The Virgo Environment Traced in CO survey (VERTICO; an ALMA Large Program) — 2019–2022
`https://sites.google.com/view/verticosurvey/home`
2. The JCMT Dust & Gas in Nearby Galaxies Legacy Survey (JINGLE) — 2017–2021
`https://www.eaobservatory.org/jcmt/science/large-programs/jingle/`

1. The South Pole Telescope Collaboration 2013–2017
   `https://pole.uchicago.edu/public/Home.html`

## Awards

| | |
|---|---|
| Visiting Scholar, Tsinghua University, Beijing, China | 2019 |
| Mitacs Globalink Research Award | 2018 |
| McMaster Graduate Fellowship | 2018 |
| McGill University Graduate Excellence Award in Physics | 2015 and 2016 |
| Carl Reinhardt Fellowship | 2015 |
| McGill and Novelis Global Technologies Summer Research Award | 2014 |
| McGill Summer Research Award | 2013 and 2015 |